
Software Requirements Specification

for

EaseShop Central Pvt Ltd Online System

Version 1.0 approved

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Revision History

Name	Date	Reason for Changes	Version
Initial Draft	2024-12-01	Document creation and initial feature specifications.	1.0
M. Ahnaf (Reviewer)	2024-12-02	Updates based on stakeholder feedback.	1.0
MN. Sihanass	2024-12-03	Adjustments to system features and inclusion of diagrams.	1.0
Tony Stark (QA Team)	2024-12-04	Added performance, safety, and non-functional requirements.	1.0
Ronaldo C. (Approver)	2024-12-05	Approved the final version with all revisions.	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document outlines the functional and non-functional requirements for the development of the EaseShop Central Digital Platform. The purpose of this project is to replace the current manual system with a modern, computerized solution designed to streamline operations, enhance customer satisfaction, and improve business efficiency. The platform will cater to both customer and administrative needs, providing an intuitive user interface and robust management tools.

1.2 Document Conventions

- **Bold Text:** Highlights critical requirements or terms.
- **TBD (To Be Determined):** Indicates areas requiring further clarification.
- **Numbered Sections:** Used for structured navigation and ease of reference.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following stakeholders:

- **Developers:** To design and implement the platform based on specified requirements.
- **Project Managers:** To align the system's development with business objectives and timelines.
- **Testers:** To validate the platform's functionality, security, and performance.
- **Business Stakeholders:** To ensure alignment with organizational goals and strategic priorities.

Readers are encouraged to begin with the **Introduction** for an overview, followed by **Overall Description** and **System Features** for detailed requirements.

1.4 Product Scope

The EaseShop Central Digital Platform is a comprehensive e-commerce solution aimed at digitalizing the shop's operations. The platform will enable customers to browse and purchase products online while equipping administrators with tools for managing inventory, orders, and customer accounts. By automating key processes, the platform will reduce operational inefficiencies, enhance data accuracy, and improve overall customer experience.

1.5 References

- Documentation of current operational processes at EaseShop Central PVT (Ltd).
- Industry best practices for e-commerce platform design.
- Regulatory guidelines for online transactions in Sri Lanka, including data protection laws.

2. Overall Description

2.1 Product Perspective

The EaseShop Central Digital Platform is a tailored e-commerce solution designed to modernize the shop's operations by transitioning from a manual system to an automated, computer-based system. It will unify inventory management, order tracking, and customer interactions into a single platform. The system will be accessible as a web-based application, with potential for mobile app integration, and will support both customer-facing and administrative functionalities.

The platform will integrate with:

- **Payment Gateways:** For secure transaction processing.
- **Customer Relationship Management (CRM) Tools:** To manage user interactions and preferences.
- **Delivery Services:** To streamline order fulfillment and tracking.

2.2 Product Functions

The core functionalities of the platform include:

1. **Customer Management:**
 - User registration, profile management, and order history tracking.
2. **Product Catalog:**
 - Browse, search, and filter products with detailed descriptions and images.
3. **Shopping Cart and Checkout:**
 - Add/remove items, apply discounts, and process secure payments.
4. **Order Management:**
 - Track orders, manage returns, and notify users of delivery status.
5. **Inventory Management:**
 - Monitor stock levels and update product availability in real-time.
6. **Reports and Analytics:**
 - Generate sales reports, customer behavior analytics, and inventory trends.

2.3 User Classes and Characteristics

The system will cater to the following user groups:

1. **Customers:**
 - **Characteristics:** Varying levels of technical proficiency, primarily seeking ease of use and secure transactions.
 - **Needs:** Intuitive interface for browsing, ordering, and payment.
2. **Administrators:**
 - **Characteristics:** Familiarity with managing data and generating reports.
 - **Needs:** Efficient tools for inventory updates, order management, and customer support.
3. **Delivery Personnel:**
 - **Characteristics:** Minimal system interaction, primarily focused on order updates.
 - **Needs:** Simple interface for updating delivery status.

2.4 Operating Environment

1. Platform:

- Web-based application compatible with modern browsers (Chrome, Edge, Firefox).

2. Server Environment:

- Linux-based hosting utilizing a LAMP stack (Linux, Apache, MySQL, PHP).

3. Client Environment:

- Devices running Windows, macOS, Android, or iOS with stable internet access.

2.5 Design and Implementation Constraints

1. Budgetary Limitations:

- Development should follow a cost-effective approach without compromising quality.

2. Performance Requirements:

- Must handle at least 500 concurrent users with minimal latency.

3. Regulatory Compliance:

- Adhere to Sri Lanka's data protection laws and e-commerce regulations.

4. Integration:

- Seamless integration with existing CRM tools and delivery services.

2.6 User Documentation

The following documentation will be provided:

1. **User Manual:** Comprehensive guide for customers and administrators.
2. **Quick Start Guide:** Simplified instructions for basic operations.
3. **FAQs:** Address common issues and troubleshooting steps.

2.7 Assumptions and Dependencies

1. **Internet Availability:** Users require a stable internet connection to access the platform.
2. **Third-Party Services:** Dependence on reliable third-party payment gateways and delivery services.
3. **User Devices:** Modern devices with updated web browsers are assumed for optimal performance.
4. **Hosting Services:** The platform assumes the availability of dependable cloud-hosted services for deployment.

3. External Interface Requirements

3.1 User Interfaces

The platform's user interfaces are designed to provide a seamless and intuitive experience for all user classes.

1. **Customer Interface:**
 - **Features:**
 - A responsive web design optimized for desktops, tablets, and mobile devices.
 - A homepage displaying featured products, a search bar, and quick links to product categories.
 - An account dashboard showing profile details, order history, and saved preferences.
 - A streamlined checkout process with multiple payment options and a progress indicator.
 - **UI Standards:**
 - Adherence to accessibility standards (e.g., WCAG 2.1).
 - Localization support for multiple languages (English, Sinhala, Tamil).

2. Administrator Interface:

- **Features:**

- Dashboard displaying key metrics such as sales, inventory status, and customer activity.
- Role-based access control with restricted features based on admin levels.
- Tools for managing products, orders, and customer accounts.

- **UI Standards:**

- Consistent input validation, tooltips, and error messages.

3. Delivery Personnel Interface:

- **Features:**

- Mobile-friendly pages for updating order delivery status.
- Integration with map services for navigation and tracking.

3.2 Hardware Interfaces

The system will interact with the following hardware components:

1. Customer Devices:

- Smartphones, tablets, desktops, and laptops with internet connectivity.

2. Server Hardware:

- Cloud-hosted servers with scalable infrastructure, including:
 - Minimum 16 GB RAM and quad-core processors.
 - High-performance SSD storage for rapid data retrieval.

3. Point-of-Sale (POS) Terminals (Optional):

- Physical devices for processing in-store payments, integrated via APIs.

3.3 Software Interfaces

The platform relies on and integrates with various software systems:

1. **Web Browsers:**
 - Compatibility with Chrome (v100+), Edge (v95+), Firefox (v90+), and Safari (v14+).
2. **Payment Gateways:**
 - Integration with secure services like PayHere or DirectPay for local transactions.
3. **Delivery Service APIs:**
 - Support for logistics platforms such as local courier services for tracking and fulfillment.
4. **Database Management System:**
 - MySQL for backend operations to manage and store data efficiently.
5. **CRM Software:**
 - Compatibility with tools like HubSpot or Salesforce for managing customer relationships.

3.4 Communications Interfaces

The platform will support the following communication protocols and methods:

1. **HTTP/HTTPS:**
 - All interactions will be secured using HTTPS for encrypted communication.
2. **SMTP:**
 - Integrated email notifications for account verification, order confirmations, and updates.
3. **REST APIs:**
 - Interfaces for integration with third-party systems such as payment gateways and delivery services.
4. **WebSockets (Optional):**
 - Real-time updates for stock changes, order statuses, or notifications.
5. **Mobile Network Protocols:**
 - Support for mobile app integration using Wi-Fi or cellular networks (3G, 4G, 5G).

4. System Features

This section outlines the platform's key features, describing their purpose, functionality, and associated requirements.

4.1 Customer Registration and Login

4.1.1 Description and Priority

The system must allow customers to securely register and log in to their accounts. This feature is of high priority, ensuring secure and personalized access to user-specific data such as order history, saved addresses, and preferences.

4.1.2 Stimulus/Response Sequences

- **Stimulus 1:** Customer enters registration details (username, email, password).
 - **Response:** System validates details (e.g., unique username, valid email format) and creates an account upon successful validation.
- **Stimulus 2:** Customer submits login credentials.
 - **Response:** System verifies credentials and grants access if valid, or displays an error message for invalid attempts.

4.1.3 Functional Requirements

- REQ-1: Allow customers to create accounts with unique credentials.
- REQ-2: Encrypt passwords in the database.
- REQ-3: Validate email and password formats during registration.
- REQ-4: Provide error messages for failed login attempts.

4.2 Product Browsing and Selection

4.2.1 Description and Priority

Customers must be able to browse products by category, view details, and add items to the shopping cart. This feature is high priority as it directly impacts user interaction and sales.

4.2.2 Stimulus/Response Sequences

- **Stimulus 1:** Customer selects a product category.
 - **Response:** System displays relevant products with images, descriptions, and prices.
- **Stimulus 2:** Customer adds a product to the cart.
 - **Response:** System updates the cart and provides a summary, including item count and total cost.

4.2.3 Functional Requirements

- REQ-1: Display products with detailed information, including images and availability.
- REQ-2: Enable users to add multiple products to the cart.
- REQ-3: Dynamically update cart summaries in real time.

4.3 Order Processing and Checkout

4.3.1 Description and Priority

The system must facilitate a smooth checkout process, enabling customers to review orders, enter shipping details, and make payments. This is a high-priority feature as it is integral to completing purchases.

4.3.2 Stimulus/Response Sequences

- **Stimulus 1:** Customer proceeds to checkout.
 - **Response:** System displays order details, total cost, and shipping options.
- **Stimulus 2:** Customer enters shipping details and confirms payment.
 - **Response:** System processes the payment and displays an order confirmation.

4.3.3 Functional Requirements

- REQ-1: Provide an order summary at checkout.
- REQ-2: Validate shipping and payment information.
- REQ-3: Integrate secure payment gateways for transactions.
- REQ-4: Send an email confirmation after successful order placement.

4.4 Inventory Management

4.4.1 Description and Priority

Admins must be able to manage inventory, monitor stock levels, and update product availability. This feature is high priority for ensuring smooth operations.

4.4.2 Stimulus/Response Sequences

- **Stimulus 1:** Admin updates stock levels for a product.
 - **Response:** System reflects the updated stock in real time.
- **Stimulus 2:** Stock levels drop below a threshold.
 - **Response:** System sends a low-stock alert to the admin.

4.4.3 Functional Requirements

- REQ-1: Allow admins to add, update, or remove inventory items.
- REQ-2: Display real-time stock levels.
- REQ-3: Trigger low-stock alerts.

4.5 Reports and Analytics

4.5.1 Description and Priority

Admins must be able to generate reports on sales, inventory, and customer behavior. This feature is medium priority as it supports business decision-making.

4.5.2 Stimulus/Response Sequences

- **Stimulus 1:** Admin requests a sales report for a specific date range.
 - **Response:** System generates and displays the report.

4.5.3 Functional Requirements

- REQ-1: Generate detailed sales and inventory reports.
- REQ-2: Provide export options for reports (e.g., CSV, PDF).
- REQ-3: Enable data visualization through charts and graphs.

4.6 Order Tracking

4.6.1 Description and Priority

Customers must be able to track their orders in real time. This is a medium priority feature as it enhances transparency and customer satisfaction.

4.6.2 Stimulus/Response Sequences

- **Stimulus 1:** Customer views the "My Orders" section.
 - **Response:** System displays current order status and tracking information.

4.6.3 Functional Requirements

- REQ-1: Display current order status for all purchases.
- REQ-2: Integrate with third-party tracking systems for real-time updates.

Note: Diagrams related to this document attached outside of the document.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

1. **System Responsiveness:**

- The platform must respond to user interactions, such as clicks and searches, within **2 seconds** under normal load conditions.

2. **Concurrent Users:**

- The system should support up to **500 concurrent users** without performance degradation and scale to accommodate **10,000 users** during peak periods.

3. **Data Processing:**

- Reports and analytics must be generated within **30 seconds** for datasets of up to **1 million records**.

4. **Scalability:**

- The platform should enable horizontal scaling to accommodate growing user and data demands.

5.2 Safety Requirements

1. **Data Integrity:**

- The platform must ensure data consistency and prevent loss or corruption during unexpected shutdowns or system crashes.

2. **Backup and Recovery:**

- Daily automated backups must be implemented, with recovery options available to restore data within **2 hours** in case of failure.

3. **Error Handling:**

- The system must display informative error messages to users while logging errors internally for analysis and resolution.

4. **User Protection:**

- Critical operations, such as data deletion or order cancellation, must include confirmation prompts to avoid accidental actions.

5.3 Security Requirements

1. Authentication:

- Multi-factor authentication (MFA) must be implemented for admin accounts, with optional MFA for customers.

2. Data Encryption:

- Sensitive data, such as passwords and payment information, must be encrypted both in transit (using SSL/TLS) and at rest (using AES-256 encryption).

3. Access Control:

- Role-based access control (RBAC) must restrict access to authorized features based on user roles (e.g., admin, customer).

4. Audit Logs:

- The platform should maintain detailed logs of user and admin activities for auditing purposes.

5. Compliance:

- The system must adhere to Sri Lanka's **Personal Data Protection Act No. 9 of 2022** and **PCI DSS** for secure payment handling.

5.4 Software Quality Attributes

1. Availability:

- The platform must maintain an uptime of at least **99.9%** annually, ensuring reliable access for users.

2. Usability:

- The user interface must be intuitive and adhere to established design standards to accommodate users of varying technical proficiency.

3. Maintainability:

- The platform's codebase should follow modular design principles, facilitating easier updates, debugging, and feature enhancements.

4. Reliability:

- The system should ensure error-free operations for at least **10 consecutive hours** under normal conditions.

5.5 Business Rules

1. Discount Policies:

- Discounts can only be applied during admin-defined promotional periods.

2. Tax Calculations:

- Taxes will be calculated dynamically based on the user's location and local regulations.

3. Returns and Refunds:

- Customers can initiate returns within **30 days** of purchase, subject to terms specified by the shop.

4. Loyalty Program:

- Registered users will earn points for purchases, redeemable for discounts.

5. Data Retention:

- Customer data will be retained for **5 years**, after which it will be archived or deleted in compliance with regulations.

6. Other Requirements

6.1 Legal and Compliance Requirements

The system must comply with the following legal and regulatory frameworks:

1. Sri Lanka Personal Data Protection Act No. 9 of 2022:

- Ensure all customer data is handled securely, with explicit consent for data collection and usage.

2. E-Commerce Regulations:

- Adhere to local laws for online transactions, ensuring transparency in product pricing and terms of service.

3. PCI DSS Compliance:

- Payment processing must meet security standards for safeguarding financial data.

4. Consumer Affairs Authority Act No. 9 of 2003:

- Protect consumer rights with clear policies for refunds, returns, and customer support.

6.2 Localization and Internationalization

The system must support localization to cater to the diverse customer base in Sri Lanka:

1. Language Support:

- Provide interfaces in English, Sinhala, and Tamil.

2. Regional Formats:

- Adapt to local formats for dates, currencies (LKR), and measurements.

6.3 Ethical Considerations

The platform must adhere to ethical guidelines:

1. Data Privacy:

- Limit data collection to what is necessary for operations and ensure it is not misused.

2. Transparency:

- Inform users about data usage and provide accessible privacy and terms of service policies.

3. Fair Practices:

- Avoid misleading information about products, services, or pricing.

6.4 Environmental Impact

The system should support environmentally sustainable practices:

1. Paperless Operations:

- Transition to digital invoices and receipts to minimize paper usage.

2. Energy Efficiency:

- Optimize server operations to reduce energy consumption.

7. Appendix A: Glossary

This section defines specialized terms and abbreviations used in the document for clarity among stakeholders.

Term	Definition
MFA (Multi-Factor Authentication)	A security process requiring multiple verification methods for user authentication.
PCI DSS	Payment Card Industry Data Security Standard, ensuring secure handling of card information.
RBAC (Role-Based Access Control)	A method for restricting system access based on user roles and permissions.
ER Diagram	Entity-Relationship Diagram, which represents the data model of the system.
DFD (Data Flow Diagram)	Visual representation of the flow of information within the system.
SLA (Service Level Agreement)	A contract that defines the expected performance and availability standards for the system.

8. Appendix B: Analysis Models

This section contains the key diagrams provided for the analysis and modeling of the EaseShop Central Digital Platform.

1. Entity-Relationship (ER) Diagram

The ER diagram provides a detailed view of the system's data structure, including relationships between key entities like customers, orders, products, and inquiries.

2. Use Case Diagram

The use case diagram outlines the interactions between system users (customers and admins) and the system, showing the primary functions such as customer management, product management, and order processing.

3. Activity Diagram (Swimlane)

The activity diagram demonstrates the workflows for key operations, such as browsing products, adding items to the cart, placing orders, and managing inquiries.

4. Data Flow Diagram (DFD)

- **DFD Level 0:** High-level overview of the system processes, including customer management, product management, and order processing.
- **DFD Level 1 and Level 2:** Detailed breakdowns of data exchanges, showing how inputs like customer information are processed into outputs like orders and reports.

5. Class Diagram

The class diagram provides an object-oriented view of the system, showing the attributes and methods of key components such as Customer, Admin, Order, Cart, and Product.

(These diagrams are attached as visual aids and serve as a comprehensive guide to the system's design and flow.)

9. Appendix C: To Be Determined List

This section lists pending decisions or incomplete items that require resolution during the project lifecycle.

Item	Description	Expected Resolution Date
Third-Party API Selection	Finalize APIs for payment gateways and delivery integrations.	December 17, 2024
Language Support	Confirm additional languages (e.g., Tamil) for platform localization.	February 20, 2025
Hosting Provider	Select a cloud hosting provider (e.g., AWS, Azure) for deployment.	November 16, 2024
System Integration Tools	Decide on tools for CI/CD pipeline integration for streamlined updates.	January 25, 2025